GMI German-Malaysian Institute

DIPLOMA PROGRAMME

LAB WORK 5

Academic Period: January 2026

Program	:	**CYBER SECURITY TECHNOLOGY (CBS)**
Course	:	**NETWORK MANAGEMENT & MONITORING SYSTEMS**
Code	:	**NTC 1052**
Total Marks	:	**20 MARKS**

Title	:	**CREATING CUSTOM LOGO HOSTS**

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Introduction

This lab guides you through creating and applying custom logos for hosts in Nagios on an openSUSE virtual machine to enhance visualization in the status map.

1 Lab 6: Creating Custom Logo for Hosts

1.1 Objective

Create and apply custom logos for hosts in Nagios to enhance visualization in the status map.

1.2 Requirements

- openSUSE virtual machine with Nagios installed
- Administrative access (root or sudo privileges)
- Image editing tool (e.g., gimp, installed via zypper install gimp)
- Image files in PNG format (40x40 pixels recommended)

1.3 Procedure

1. Log in to your openSUSE VM and open a terminal.
2. Install GIMP if not already installed:

    sudo zypper install gimp

```
root@nagios-server:~# apt update && apt install -y imagemagick
Hit:1 http://deb.debian.org/debian bookworm InRelease
Get:2 http://security.debian.org bookworm-security InRelease [34.8 kB]
Get:3 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:4 http://security.debian.org bookworm-security/main amd64 Packages [349 kB]
Get:5 http://security.debian.org bookworm-security/main Translation-en [218 kB]
Fetched 657 kB in 0s (1654 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
23 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fontconfig fonts-droid-fallback fonts-noto-mono fonts-urw-base35 ghostscript gsfonts hicolor-icon-theme
  imagemagick-6-common imagemagick-6.q16 libavahi-client3 libavahi-common-data libavahi-common3 libcairo2 libcups2
  libdatrie1 libdjvulibre-text libdjvulibre21 libfftw3-double3 libfontenc1 libfribidi0 libglib2.0-0 libglib2.0-data
  libgraphite2-3 libgs-common libgs10 libgs10-common libharfbuzz0b libice6 libidn12 libijs-0.35 libimath-3-1-29
  libjbig2dec0 libjxr-tools libjxr0 liblcms2-2 liblqr-1-0 libltdl7 libmagickcore-6.q16-6 libmagickcore-6.q16-6-extra
  libmagickwand-6.q16-6 libnetpbm11 libopenexr-3-1-30 libopenjp2-7 libpango-1.0-0 libpangocairo-1.0-0
  libpangoft2-1.0-0 libpaper-utils libpaper1 libpixman-1-0 libsm6 libthai-data libthai0 libwmflite-0.2-7
  libxcb-render0 libxcb-shm0 libxext6 libxrender1 libxt6 netpbm poppler-data shared-mime-info x11-common xdg-user-dirs
  xfonts-encodings xfonts-utils
Suggested packages:
```

I used imagemagick instead, because of a version mismatch issue in my environment.

3. Open GIMP and create a 40x40 pixel PNG image (e.g., a server icon).
4. Save the image as server1.png in /tmp.

5. Copy the image to the Nagios images directory:

sudo cp /tmp/server1.png /usr/share/nagios/html/images/logos/ 6. Edit the host

configuration file (e.g., /etc/nagios/hosts/server1.cfg):

```
root@nagios-server:~# convert -size 40x40 xc:#1e70ba -fill white -gravity center -pointsize 14 -annotate 0 'S1' /usr/loc
al/nagios/share/images/logos/server1.png
root@nagios-server:~# chown nagios:nagios /usr/local/nagios/share/images/logos/server1.png
root@nagios-server:~# chmod 644 /usr/local/nagios/share/images/logos/server1.png
root@nagios-server:~# file /usr/local/nagios/share/images/logos/server1.png
/usr/local/nagios/share/images/logos/server1.png: PNG image data, 40 x 40, 16-bit/color RGB, non-interlaced
root@nagios-server:~# nano /usr/local/nagios/etc/objects/lab_hosts.cfg
root@nagios-server:~# grep -E "icon_image|statusmap_image" /usr/local/nagios/etc/objects/lab_hosts.cfg
    icon_image              server1.png
    statusmap_image         server1.png
```

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sudo nano /etc/nagios/hosts/server1.cfg

```
root@nagios-server:~# cat /usr/local/nagios/etc/objects/lab_hosts.cfg
define host {
    use                     generic-host
    host_name               router1
    alias                   Lab Router 1
    address                 10.10.10.112
    contact_groups          admins
    max_check_attempts      1
    check_interval          1
    retry_interval          1
    check_command           check-host-alive
}

define host {
    use                     generic-host
    host_name               server1
    alias                   Lab Server 1
    address                 10.10.10.113
    contact_groups          admins
    icon_image              server1.png
    statusmap_image         server1.png
    max_check_attempts      1
    check_interval          1
    retry_interval          1
    check_command           check-host-alive
}
```

7. Add or update icon*mageand*

define host {

use generic-host

host_name server1

alias Server 1

address 192.168.1.100

icon_image server1.png

statusmap_image server1.png

}

\end{verbatim}

\item Save the file and verify the configuration:

\begin{verbatim} sudo nagios -v

/etc/nagios/nagios.cfg

\end{verbatim}

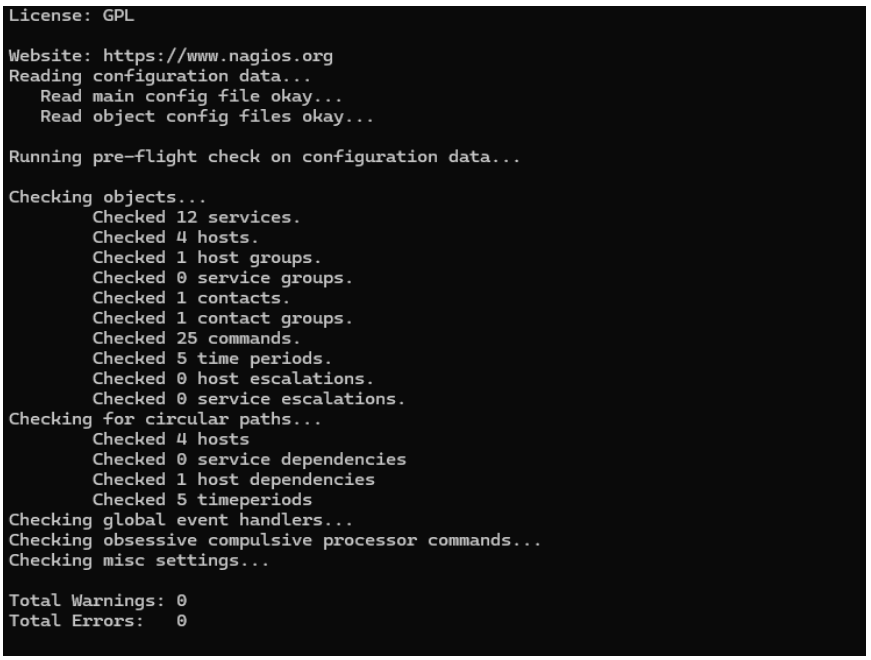
If no errors, restart Nagios:

\begin{verbatim} sudo systemctl

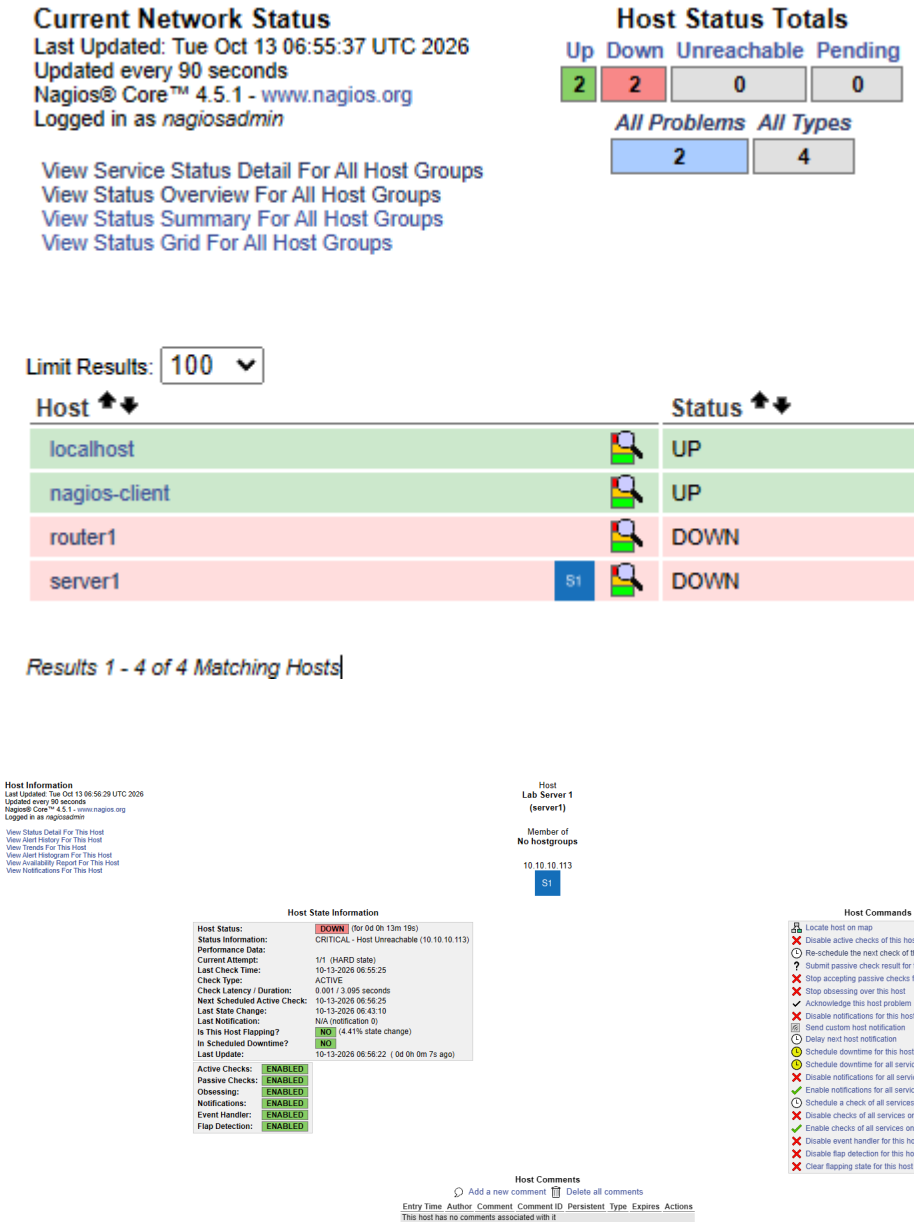
restart nagios

\end{verbatim}

\item Access the Nagios web interface (\texttt{http://<your-vm-ip>/nagios}) a



Proof of Concept :
I created a small 40x40 pixel S1 text image with a blue background for server 1 for simplicity.



1.4 Safety Notes

- Use small images (e.g., 40x40 pixels) to avoid performance issues.
- Scan images for malware before uploading.

1.5 Expected Outcomes

- Custom logo is displayed for server1 in the Nagios Status Map.
- Configuration is applied without zero error.